

Minimizing the degree of a set of real algebraic numbers, with an application: the snub cube has no constructible model

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Abstract

Given a finite set S of real algebraic numbers, how small can the largest degree of an element of $T(S)$ be made, over all affine maps $T(x) = ax + b$ with real algebraic coefficients and $a \neq 0$? This question was asked on Math StackExchange (question 2953331). We reduce it to a problem about the *invariant field* generated by the affine ratios of S , prove group-theoretic lower bounds (the Galois group of the invariant field's closure must be a section of a product of small symmetric groups), and give a linear-algebra upper-bound machine that redoes the question's worked example in a four-line computation. For target degree 2 the two sides meet: we obtain a complete, finite decision procedure. As a further consequence of the lower-bound method we prove the result that originally motivated the question: no set of points similar to a snub cube has all coordinates expressible by quadratic irrationals — indeed, none has all coordinates in the field of constructible numbers, because the snub cube possesses a similarity invariant of degree 3, namely the square of the tribonacci constant.

1 The problem

Let $S = \{s_1, \dots, s_n\}$ be a finite set of real algebraic numbers. For an algebraic number w , $\deg w = [\mathbb{Q}(w) : \mathbb{Q}]$, and $\deg S = \max_i \deg s_i$. We consider affine maps $T(x) = ax + b$ with a, b real algebraic and $a \neq 0$, and ask:

$$d(S) = \min_T \deg T(S) = \min_{a,b} \max_i \deg(as_i + b).$$

Throughout, $(p(x))_k$ denotes the k -th real root of p in increasing order, starting at $k = 0$.

Example 1 (the StackExchange example). Let

$$S = \{\alpha, \beta, \gamma\} = \{(8x^3 + 4x^2 - 10x - 37)_0, (x^6 + 4x^5 - 26x^4 - 90x^3 + 200x^2 + 100x - 775)_1, \\ (x^6 - 2x^5 - 22x^4 + 22x^3 + 142x^2 - 272x - 131)_1\} \approx \{1.74190, 4.40417, 4.88765\},$$

so $\deg S = 6$. With $a = 1$ and $b = (x^3 - 2x^2 + 5)_0 \approx -1.24190$ one gets $T(S) = \{1/2, \sqrt{10}, 1 + \sqrt{7}\}$ and $\deg T(S) = 2$, which is optimal. The example was constructed backward from its solution; the question asks for the forward direction.

*M.H. posed the problem (as Math StackExchange question 2953331), constructed its worked example — backward, which is the easy direction only in the sense that it took him and not us — and asked the snub cube question that Section 8 answers.

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An answer posted by user *mathlove* exhibits, for this specific S , a four-parameter family of optimal (a, b) , found by substituting the ansatz $as_i + b = p_i + q_i\sqrt{r_i}$ into the three minimal polynomials and eliminating by hand. We will see below (Section 7) that the family is precisely the orbit of the known solution under rational affine reparametrization and conjugate selection, and that the computation, while correct, is specific to the instance: the ansatz, the assumed degree of b , and the branch choices in the elimination are all guided by the known answer. The purpose of this note is to set up the general theory.

2 The affine reduction and the invariant field

Assume $n \geq 2$ and fix $s_1 \neq s_2$. An affine map is determined by the images $w_1 = T(s_1)$, $w_2 = T(s_2)$; writing $u = w_1$ and $v = w_2 - w_1 \neq 0$, the remaining images are

$$w_i = u + v\lambda_i, \quad \lambda_i = \frac{s_i - s_1}{s_2 - s_1} \quad (i = 3, \dots, n), \quad (1)$$

and conversely any real algebraic (u, v) with $v \neq 0$ arises from exactly one T (with $a = v/(s_2 - s_1)$, which is real and nonzero). We also set $\lambda_1 = 0$, $\lambda_2 = 1$. The numbers $\lambda_3, \dots, \lambda_n$ are a complete system of invariants of S under affine maps. Define the *invariant field*

$$K = \mathbb{Q}(\lambda_3, \dots, \lambda_n),$$

let $\tilde{K}$ be its Galois closure, and $G = \text{Gal}(\tilde{K}/\mathbb{Q})$.

Proposition 2. *For every admissible T , the field $L_T = \mathbb{Q}(w_1, \dots, w_n)$ contains K . Consequently $d(S) \geq 2$ unless all λ_i are rational, in which case $d(S) = 1$. In the other direction, $(u, v) = (0, 1)$ shows $d(S) \leq \max_i \deg \lambda_i$.*

Proof. $\lambda_i = (w_i - w_1)/(w_2 - w_1) \in L_T$. If all images were rational, all λ_i would be rational; conversely if all $\lambda_i \in \mathbb{Q}$, the map with $(u, v) = (0, 1)$ has images $0, 1, \lambda_3, \dots, \lambda_n \in \mathbb{Q}$. $\square$

K is the field of moduli of the configuration; no affine change of coordinates can shrink the joint field below it. The whole question is how the content of K can be *distributed* across the individual images: in Example 1,

$$\lambda_3 = \frac{\gamma - \alpha}{\beta - \alpha} = \frac{\frac{1}{2} + \sqrt{7}}{\sqrt{10} - \frac{1}{2}} = \frac{1 + 2\sqrt{10} + 2\sqrt{7} + 4\sqrt{70}}{39} \approx 1.18160, \quad (2)$$

a primitive element of the biquadratic field $K = \mathbb{Q}(\sqrt{7}, \sqrt{10})$ of degree 4; yet the optimum spreads K over three images of degree at most 2.

3 Galois-theoretic lower bounds

Theorem 3 (section obstruction). *If some admissible T achieves $\deg(as_i + b) \leq d_i$ for all i , then $G = \text{Gal}(\tilde{K}/\mathbb{Q})$ is a section (a quotient of a subgroup) of $S_{d_1} \times \dots \times S_{d_n}$.*

Proof. Let N_i be the Galois closure of $\mathbb{Q}(w_i)$ and $\tilde{L}$ the compositum of the N_i , which is Galois over $\mathbb{Q}$. Restriction gives an injection $\text{Gal}(\tilde{L}/\mathbb{Q}) \hookrightarrow \prod_i \text{Gal}(N_i/\mathbb{Q})$, since an automorphism trivial on every N_i is trivial on their compositum; and $\text{Gal}(N_i/\mathbb{Q})$ acts faithfully on the $\deg w_i \leq d_i$ roots of the minimal polynomial of w_i , so it embeds in S_{d_i} . By Proposition 2, $K \subseteq L_T \subseteq \tilde{L}$, hence $\tilde{K} \subseteq \tilde{L}$ and G is a quotient of $\text{Gal}(\tilde{L}/\mathbb{Q})$. $\square$

Corollary 4. $d(S) \geq$ the largest prime divisor of $|G|$.

Proof. If $\max_i d_i = d$, then $|G|$ divides $|S_d|^n = (d!)^n$, so every prime divisor of $|G|$ is at most d . $\square$

Sharper obstructions come from the structure of sections of $\prod S_{d_i}$ with all $d_i \leq 3$: such a group $B \leq \prod (S_3 \text{ or } S_2)$ has an abelian Sylow 2-subgroup of exponent 2 (contained in a conjugate of $\prod C_2$), and a normal 3-subgroup $B \cap \prod C_3$ with elementary abelian 2-quotient. Both properties pass to quotients. This settles the smallest nontrivial case completely:

Theorem 5 (quartic classification). *Let $n = 3$ and let $\lambda = \lambda_3$ have degree 4.*

- If $\mathbb{Q}(\lambda)$ is biquadratic (Galois group V_4), then $d(S) = 2$.
- In all other cases (C_4, D_4, A_4, S_4), $d(S) = 4$.

In particular $d(S) = 3$ never occurs for a quartic invariant.

Proof. Lower bounds. Suppose all three images have degree ≤ 2 . Then $L = \mathbb{Q}(w_1, w_2, w_3)$ is a compositum of at most three quadratic fields, hence a Galois extension with group an elementary abelian 2-group; $L \supseteq \mathbb{Q}(\lambda)$ forces $L \supseteq \tilde{K}$, so G is a quotient of an exponent-2 abelian group. This eliminates C_4 (exponent), and D_4, A_4, S_4 (nonabelian). Suppose instead all three images have degree ≤ 3 ; by Theorem 3, G is a section of a product of copies of S_3 and S_2 . For $G \in \{C_4, D_4, S_4\}$ the Sylow 2-subgroup (C_4, D_4, D_4 respectively) is not a section of an elementary abelian 2-group, either by exponent or by nonabelianness. For $G = A_4$: the image of the normal 3-subgroup of B is a normal 3-subgroup of A_4 , hence trivial (A_4 's proper normal subgroups are 1 and the 2-group V_4), so A_4 would be a quotient of a subgroup of an elementary abelian 2-group — absurd. Hence in the non- V_4 cases $d(S) \geq 4$, and $(u, v) = (0, 1)$ attains 4.

Upper bound for V_4 . Let F_1, F_2, F_3 be the three quadratic subfields of the biquadratic field $M = \mathbb{Q}(\lambda)$. The condition

$$w_3 - w_1 = \lambda(w_2 - w_1), \quad w_i \in F_i,$$

is a system of 4 $\mathbb{Q}$ -linear equations (one per basis component of M) in $2 + 2 + 2 = 6$ rational unknowns, so its solution space has dimension ≥ 2 . A solution with $w_2 = w_1$ forces $w_3 = w_1$ and $w_1 = w_2 \in F_1 \cap F_2 = \mathbb{Q}$, so the degenerate solutions form only the 1-dimensional line $w_1 = w_2 = w_3 \in \mathbb{Q}$. Any nondegenerate solution yields T with all images of degree ≤ 2 ; since $\lambda \notin \mathbb{Q}$, at least one image is irrational, and $d(S) = 2$ exactly. $\square$

Remark 6. The A_4 case shows that the section theorem is strictly stronger than naive degree bookkeeping: an A_4 -quartic field has no quadratic subfield, so *every element* of $\mathbb{Q}(\lambda)$ has degree 1 or 4 — all powers of two — yet degree 2 (and even 3) is unreachable. Divisibility of degrees alone would not detect this.

4 The upper-bound machine, and Example 1 in four lines

Let M be the Galois closure of K and $G = \text{Gal}(M/\mathbb{Q})$ (assume for the moment that images may be sought in subfields of M ; Section 6 discusses when this loses nothing). Subfields $F \subseteq M$ with $[F : \mathbb{Q}] \leq d$ correspond to subgroups of G of index $\leq d$. The machine, for a target d :

1. Enumerate assignments $(F_1, \dots, F_n)$ of such subfields, one per image.
2. Solve the $\mathbb{Q}$ -linear system $\{w_i - w_1 - \lambda_i(w_2 - w_1) = 0 \ (i = 3..n), \ w_i \in F_i\}$, adding, if $M \not\subseteq \mathbb{R}$, the linear constraint that w_1 and $w_2 - w_1$ be fixed by complex conjugation.
3. Any solution with $w_2 \neq w_1$ certifies $d(S) \leq d$.

For Example 1: $M = K = \mathbb{Q}(\sqrt{7}, \sqrt{10})$, and we try $F_1 = \mathbb{Q}$, $F_2 = \mathbb{Q}(\sqrt{10})$, $F_3 = \mathbb{Q}(\sqrt{7})$, i.e. $w_1 = a$, $w_2 = b_0 + b_1\sqrt{10}$, $w_3 = c_0 + c_1\sqrt{7}$. With $\beta = b_0 - a$ and λ from (2), expanding $w_3 - w_1 = \lambda(w_2 - w_1)$ over the basis $\{1, \sqrt{10}, \sqrt{7}, \sqrt{70}\}$ gives

$$\sqrt{10}: 2\beta + b_1 = 0, \quad \sqrt{70}: 4\beta + 2b_1 = 0, \quad 1: c_0 - a = \frac{1}{39}(\beta + 20b_1), \quad \sqrt{7}: c_1 = \frac{1}{39}(2\beta + 40b_1).$$

The $\sqrt{70}$ equation is a multiple of the $\sqrt{10}$ equation (a special feature of this λ , which is what allows a *rational* first image), leaving $b_1 = -2\beta$, $c_0 = a - \beta$, $c_1 = -2\beta$ with a, β free:

$$(w_1, w_2, w_3) = (a, a + \beta - 2\beta\sqrt{10}, a - \beta - 2\beta\sqrt{7}).$$

At $(a, \beta) = (\frac{1}{2}, -\frac{1}{2})$ this is $(\frac{1}{2}, \sqrt{10}, 1 + \sqrt{7})$ — the known optimum — and the two free parameters are exactly composition with rational affine maps. This replaces the sextic elimination of the posted answer by a 4×6 linear system, and Theorem 5 supplies the missing optimality proof.

5 Degree 2 is completely decidable

For target degree 2 the machine is *complete*, with no assumption:

Theorem 7. *$d(S) \leq 2$ if and only if all λ_i lie in a single multiquadratic field (equivalently, K is multiquadratic; then $M = K$), and for some assignment of each $F_i \in \{\mathbb{Q}\} \cup \{\text{quadratic subfields of } M\}$ the linear system of Section 4 has a solution with $w_2 \neq w_1$. Both conditions are decidable by finite rational linear algebra.*

Proof. $(\Leftarrow)$ is Section 4; reality is automatic since $K \subseteq \mathbb{R}$ and a real multiquadratic field is its own Galois closure. $(\Rightarrow)$: suppose (u, v) , $v \neq 0$, achieves images w_i of degree ≤ 2 . Then $L = \mathbb{Q}(w_1, \dots, w_n)$ is a compositum of quadratic fields: a Galois extension $\mathbb{Q}(\sqrt{r_1}, \dots, \sqrt{r_k})$, $r_j \in \mathbb{Q}$, with group C_2^k . All $\lambda_i \in L$, so K and $M \subseteq L$ are multiquadratic. It remains to move the images into subfields of M . Let $H = \text{Gal}(L/M) \leq C_2^k$ and decompose $L = \bigoplus_{\chi} L_{\chi}$ into H -eigenspaces over M ; because L is multiquadratic over $\mathbb{Q}$, each nonzero L_{χ} can be written $M \cdot \sqrt{e_{\chi}}$ with a *rational* representative e_{χ} (a product of the r_j). Pick χ with the χ -component of v nonzero, and let $\psi: L \rightarrow M$ extract the χ -coordinate: $\psi(z) = z_{\chi} / \sqrt{e_{\chi}}$. Then ψ is M -linear, so applying it to (1) gives a new admissible pair $(\psi(u), \psi(v))$ with $\psi(v) \neq 0$ and images $\psi(w_i)$. Writing $w_i = a_i + b_i\sqrt{d_i}$ with $a_i, b_i, d_i \in \mathbb{Q}$: if $\sqrt{d_i}$ has character χ , then $\psi(w_i) = b_i\sqrt{d_i} / \sqrt{e_{\chi}}$, an element of M whose square is the rational $b_i^2 d_i / e_{\chi}$, hence of degree ≤ 2 ; if not (and χ is nontrivial), $\psi(w_i) = 0$; and for trivial χ , $\psi(w_i) \in \mathbb{Q}(\sqrt{d_i}) \cap M$. In every case the new images lie in quadratic-or-rational subfields of M . $\square$

6 The general algorithm and the descent question

For general d the machine of Section 4 is a true algorithm — finitely many subgroup assignments, each a rational kernel computation — and it always yields upper bounds, while Theorem 3 yields lower bounds. Its *completeness* rests on:

Question 8 (descent). If some real algebraic (u, v) , $v \neq 0$, achieves $\deg(u + v\lambda_i) \leq d_i$ for all i , do there exist $(u', v') \in M^2$, $v' \neq 0$, achieving the same bounds?

We can prove this in two regimes beyond $d = 2$ (Theorem 7):

Lemma 9 (forced compositum). *If $\prod_i d_i < 2[M : \mathbb{Q}]$, descent holds: $L = \mathbb{Q}(w_1, \dots, w_n)$ satisfies $M \subseteq L$ and $[L : \mathbb{Q}] \leq \prod_i d_i < 2[M : \mathbb{Q}]$, so $L = M$ already.*

Lemma 10 (trace descent). *Let $E \supseteq M$ be Galois over $\mathbb{Q}$ containing u, v , let $H = \text{Gal}(E/M)$, and $S_i = \text{Stab}(w_i)$ in $\text{Gal}(E/\mathbb{Q})$. If $\text{Tr}_{E/M}(cv) \neq 0$ for some c in the fixed field of $\langle S_1, \dots, S_n \rangle$, then descent holds: the pair $(\text{Tr}_{E/M}(cu), \text{Tr}_{E/M}(cv))$ lies in M^2 and, because H is normal and fixes every λ_i , its images $\text{Tr}_{E/M}(cw_i)$ are fixed by the image of S_i in $\text{Gal}(M/\mathbb{Q})$, hence have degree $\leq [\text{Gal}(E/\mathbb{Q}) : S_i] \leq d_i$.*

The gap is that the twisted trace can annihilate v ; for junk extensions LM/M of exponent 2 the eigenspace projection of Theorem 7 repairs this (quadratic characters are rational), but for cubic junk only the plain trace is $\mathbb{Q}$ -rationally equivariant, and we do not know how to rule out its vanishing. In every worked example the question is vacuous (Lemma 9 applies) or resolved, and we conjecture descent holds in general; settling it would make the algorithm unconditionally complete. Independently of descent, upper and lower bounds often meet, as in Theorem 5, and the minimization of any monotone function of the degree profile (maximum, sum, lcm, ...) uses the same enumeration.

7 Assessment of the posted answer

In this framework, mathlove's computation is the F -assignment $(\mathbb{Q}, \mathbb{Q}(\sqrt{10}), \mathbb{Q}(\sqrt{7}))$ instance of Section 4, executed by polynomial elimination instead of linear algebra. The ansatz (quadratic images, cubic b) and the branch selections in the elimination are imported from the known solution rather than derived, other branches of the case tree are not pursued, and optimality is not addressed. The four parameters (s, t, x, y) of the answer's family are the two rational-affine degrees of freedom (a, β) above together with two conjugate/root-index selections. So the answer is a correct verification of the instance, not a general method — the general method is the subject of this note.

8 Higher dimensions and the snub cube

For configurations $P = \{p_1, \dots, p_n\} \subset \mathbb{R}^m$ up to *similarity* (the higher-dimensional version of the original question), the role of the λ_i is played by the normalized Gram data $\langle p_i - p_1, p_j - p_1 \rangle / |p_2 - p_1|^2$, and the analogue of Proposition 2 holds with the same proof: every similarity invariant of this kind lies in the field generated by the coordinates of *any* similar copy. In particular:

Proposition 11. *If some ratio of squared distances of P has degree that is not a power of 2, then no similar copy of P has all coordinates in the field of constructible numbers. (Constructible numbers lie in towers of quadratic extensions; finitely many of them generate a field of 2-power degree, and the invariant would be an element of it, so its degree would divide a power of 2.)*

This applies to the snub cube. Let t be the *tribonacci constant*, the real root of

$$t^3 = t^2 + t + 1, \quad t \approx 1.83928675521416,$$

irreducible over $\mathbb{Q}$ (no rational root), so $\deg t = 3$. A standard model of the snub cube has as vertices the 12 even permutations of $(1, 1/t, t)$ bearing an even number of minus signs together with the 12 odd permutations bearing an odd number (the mirror image swaps the parities; since reflections are similarities, chirality plays no role below). Note $1/t = t^2 - t - 1$, so all coordinates lie in the cubic field $\mathbb{Q}(t)$.

Theorem 12. *Let $P \subset \mathbb{R}^3$ be any set of points similar to the snub cube, and let L be the field generated by its coordinates. Then $3 \mid [L : \mathbb{Q}]$. Consequently the coordinates of a snub cube can never all be constructible; a fortiori they cannot all be quadratic irrationals or rationals, nor all lie in a multiquadratic field.*

Proof. Work first in the standard model and put $s = 1/t$. Take the three vertices

$$v = (1, s, t), \quad w = (s, t, 1), \quad x = (-s, -t, 1),$$

(all lie in the even-permutation, even-sign class). Using $st = 1$, $s + t = t^2 - 1$, and $s^2 + t^2 = (s + t)^2 - 2st = t^4 - 2t^2 - 1 = 2t$ (by $t^4 = 2t^2 + 2t + 1$):

$$|v - w|^2 = (1 - s)^2 + (s - t)^2 + (t - 1)^2 = 2(1 + s^2 + t^2 - s - t - st) = 2(1 + 2t - t^2),$$

$$|v - x|^2 = (1 + s)^2 + (s + t)^2 + (t - 1)^2 = 2(1 + st) + 2(s^2 + t^2) + 2s - 2t = 4 + 2(s + t) = 2(t^2 + 1).$$

Their ratio is

$$\rho = \frac{|v - x|^2}{|v - w|^2} = \frac{t^2 + 1}{1 + 2t - t^2} = t^2,$$

since $t^2(1 + 2t - t^2) = t^2 + 2t^3 - t^4 = t^2 + 2(t^2 + t + 1) - (2t^2 + 2t + 1) = t^2 + 1$. (Numerically, $\rho = 8.76595\dots/2.59120\dots = 3.38298\dots = t^2$.) Now $t^2 \notin \mathbb{Q}$ (else $\deg t \leq 2$), and $\mathbb{Q}(t^2) \subseteq \mathbb{Q}(t)$ has degree dividing the prime 3, so $\deg \rho = 3$.

If P is similar to the snub cube, the similarity carries v, w, x to vertices v', w', x' of P and preserves ratios of squared distances, so $\rho = |v' - x'|^2/|v' - w'|^2 \in L$. Hence $\mathbb{Q}(\rho) \subseteq L$ and $3 = [\mathbb{Q}(\rho) : \mathbb{Q}]$ divides $[L : \mathbb{Q}]$. A field generated by finitely many constructible numbers has degree 2^k , and $3 \nmid 2^k$. $\square$

Remark 13. The same argument, with the same three-line computation, shows the snub cube cannot be realized with coordinates in *any* field of 2-power degree. It is the polyhedral analogue of the non-constructibility of the regular heptagon. By contrast, the regular icosahedron and dodecahedron have models with every coordinate in $\mathbb{Q}(\sqrt{5})$, and so are constructible. The snub dodecahedron is amenable to the same method: its coordinate field is a cubic extension of $\mathbb{Q}(\sqrt{5})$, and the analogous distance-ratio computation produces an invariant of degree divisible by 3.

Remark 14 (limits of the method in higher dimensions). The lower-bound side transfers to $\mathbb{R}^m$ wholesale, but the upper-bound side does not: rational similarity invariants do *not* guarantee rational coordinates. An equilateral triangle has all invariants rational, admits rational coordinates in $\mathbb{R}^3$ (the unit basis vectors), but none in $\mathbb{R}^2$ (a lattice triangle has rational area, while an equilateral triangle with rational side lengths squared has area $\sqrt{3} \cdot (\text{rational})$). Realizing a rational Gram matrix by low-degree coordinates is a question about rational congruence of quadratic forms (Hasse–Minkowski), and by Mäev-type universality, realization spaces of polytopes can force invariant fields of arbitrarily high degree, so hard instances are unavoidable. The 1-dimensional theory of Sections 2–5 is the part that admits a clean algorithm.

9 Implementation notes

The question’s $(p(x))_k$ notation is exactly the representation used by FLINT’s `qqbar` module (minimal polynomial plus isolated root enclosure), which Malachite plans to port. The 1-dimensional algorithm decomposes into:

- **Invariants:** λ_i by `qqbar` arithmetic (resultant-based field operations with enclosure refinement).
- **Fields:** a primitive element for K ; minimal polynomials require univariate factorization over $\mathbb{Q}$ (van Hoeij), which is also the engine for the subfield algorithm of van Hoeij–Klüners–Novocin, which enumerates all subfields of a number field in polynomial time — precisely the F_i enumeration for small d . Galois closures and `Gal` for the lower bounds are available in Pari/GP for moderate degrees.
- **Linear algebra:** the systems of Section 4 are exact rational kernel computations; Malachite’s `Rational` matrices suffice today.
- **Degree 2:** Theorem 7 needs no Galois machinery at all — only square-class bookkeeping in a multiquadratic field — and would make a good first target: it already decides questions like “does this configuration have a model over quadratic irrationals?”, the question answered negatively for the snub cube above.

Source. The original question is Math StackExchange 2953331, “Minimizing the degree of a set of real algebraic numbers” (2018); the posted answer discussed in Section 7 is by user mathlove.